

Orthogonal Scores, Regularization Bias and Treatment-Effect Heterogeneity

A reproducible causal-ML and staggered-DiD laboratory

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<https://github.com/hakangulmez/causal-ml-lab>

Abstract

We ask when orthogonal scores improve treatment-effect estimation relative to regularized plug-in methods. We compare two Monte Carlo designs and apply partially linear, interactive-regression and staggered-adoption estimators to retirement-account and county-employment examples. In the sparse simulation, plug-in bias is 0.610 versus 0.003 for cross-fitted double machine learning, whose coverage is 94 percent. In the nonlinear design, the in-sample orthogonal comparator has bias 0.007, showing that sample splitting does not imply a universal finite-sample ranking. The retirement-account random-forest ATE is \$8172. Orthogonality reduces nuisance-estimation sensitivity but does not establish unconfoundedness, parallel trends or policy transportability.

1 Introduction and contribution

Prediction quality and causal identification solve different problems. Flexible prediction can approximate a conditional mean while regularization induces substantial first-order bias in a treatment coefficient. Conversely, a favorable finite-sample performance comparison does not prove that cross-fitting universally improves precision. The empirical question here is when orthogonal scores help in declared designs and what they can establish in observational applications.

Chernozhukov et al. (2018) combine Neyman-orthogonal scores with sample splitting; Bach et al. (2022) provide the software implementation used in the empirical replication. Callaway and Sant’Anna (2021) define group-time treatment effects under staggered adoption; Sun and Abraham (2021) address heterogeneous event-study effects; Goodman-Bacon (2021) decomposes conventional fixed-effects comparisons. Rambachan and Roth (2023) formalize sensitivity to departures from parallel trends. Wager and Athey (2018) develop forest-based heterogeneous-effect inference.

The contribution is an auditable laboratory joining estimands, implemented moments and diagnostic accounting. It preserves a nonlinear simulation in which the historic in-sample orthogonal estimator performs well, and adds a predeclared sparse design specifically to illustrate regularization-bias theory, not to tune outcomes. The applications distinguish partial-linear coefficients from ATEs, eligibility from participation, and descriptive forest averages from formally inferred held-out group effects. The paper reports failures and scope cuts alongside successful estimates.

Table 1: Analysis inputs and estimand-defining variables

Variable	Source / ID	Transformation	Sample
Y: net financial assets	Public SIPP extract; net_tfa	Level, historical USD	9915 persons
D: eligibility	Same extract; e401	Binary; not participation	Same sample
X: continuous	age, inc, educ, fsize	Levels; train-fold scaling for linear learner	Same sample
X: indicators	marr, twoearn, db, pira, hown	As supplied	Same sample
Panel outcome	County panel; lemp	Log teen employment	2003–2007
Adoption / adjustment	mpdta: first.treat, lpop	Cohort year / log population	2500 county-year rows
Simulation variables	Author-constructed simulations	Gaussian confounding; known potential outcomes	Seeded Monte Carlo designs
Dose example	Author-constructed dose design	Dose-specific effect and derivative	Original synthetic draw

Sources: results/manifest.json, r_manifest.json, pension.csv, mpdta_group_time.csv. Historical SIPP measurements refer to the extract described in DATA.md; no raw rows are included.

2 Data

2.1 Historical retirement-account extract

We obtain the public SIPP extract at runtime and retain it privately.¹ The 1991 extract contains measurements referring to 1990; the outcome is net financial assets, and the treatment is eligibility rather than realized participation. We use 9915 observations and the raw covariates in the table, with no winsorization, income filtering, polynomial expansion or missing-row deletion. We preserve native data types when reading cached observations to maintain numerical consistency.

Access for analysis does not establish a separate dataset-redistribution licence. Only aggregate outputs enter this document. We record source provenance, licence scope and software versions in the supplementary documentation. No raw microdata, row-level predictions or private training caches are embedded.

2.2 Panel and synthetic inputs

We use the county teen-employment panel distributed with the staggered-adoption software.² The analysis includes 2500 county-year rows over 2003–2007. The population-adjusted group-time design and the unadjusted TWFE design are intentionally distinguished. We construct seeded simulation demonstrations; their potential-outcome functions and nuisance settings are documented below and in the appendix. They are illustrative designs rather than calibrations to a population of policy decisions.

3 Econometric model

3.1 Potential outcomes and staggered-adoption estimands

For binary eligibility, let $Y_i(1)$ and $Y_i(0)$ be potential financial-asset outcomes and D_i observed eligibility. Consistency implies $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. IRM identification requires conditional independence $(Y_i(1), Y_i(0)) \perp D_i \mid X_i$ and overlap $0 < m_0(X_i) < 1$. The target is $\theta_{ATE} =$

¹Acquisition uses the DoubleML data fetcher; source provenance and the distinction between analysis and redistribution are documented in DATA.md.

²The panel is `mpdta` in the pinned R `did` package.

$\mathbb{E}[Y_i(1) - Y_i(0)]$. These assumptions cannot be established from fit quality or a positive coefficient. For a panel, let G_i be first treatment time, with $G_i = 0$ denoting never treated; $D_{it} = 1\{G_i > 0, t \geq G_i\}$. Under consistency, no anticipation and suitable conditional parallel trends, the group-time estimand is

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g], \quad t \geq g. \quad (1)$$

The synthetic target averages effects over treated person-time cells. The simple aggregated estimator weights each post-treatment cohort-time effect by cohort size:

$$\theta^{\text{simple}} = \sum_{g>0} \sum_{t \geq g} w_{gt} ATT(g, t), \quad w_{gt} = \frac{n_g}{\sum_{g'>0} \sum_{t' \geq g'} n_{g'}}. \quad (2)$$

For the unconditional estimator, we compare outcome changes from $g-1$ to t against never-treated units. For the primary group-time estimator, we use never-treated controls, a universal base period and analytic influence inference, followed by simple aggregation.³ Population enters only in the county application. Dynamic aggregation averages cohort effects at common event time, with cohort weights adjusted to available cells.

For the conditional group-time estimator, we write $\Delta Y = Y_t - Y_{g-1}$, $G_g = 1(G = g)$, $G_0 = 1(G = 0)$, $p_g(X) = \Pr(G = g \mid G \in \{g, 0\}, X)$ and $m_{0,g}(X) = \mathbb{E}[\Delta Y \mid G = 0, X]$. Our doubly robust comparison has the normalized-weight moment

$$ATT(g, t) = \mathbb{E} \left[\left\{ \frac{G_g}{\mathbb{E}G_g} - \frac{p_g(X)G_0/(1-p_g(X))}{\mathbb{E}[p_g(X)G_0/(1-p_g(X))]} \right\} (\Delta Y - m_{0,g}(X)) \right]. \quad (3)$$

The county fit conditions on log population; synthetic fits use the intercept-only version. We use the nuisance-fitting and aggregation settings documented in the appendix.

The TWFE model is $Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + e_{it}$. We use two-way within residualization for the balanced-panel estimator and unit-cluster covariance for the R fixed-effects estimator.⁴ We estimate Sun–Abraham cohort-by-event interactions and aggregate the post-treatment coefficients.⁵ TWFE need not recover the treated-cell target under heterogeneous dynamics because its comparisons can use already-treated units.

3.2 PLR and IRM moment conditions; cross-fitting

The partially linear model is

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \quad \mathbb{E}[U_i \mid D_i, X_i] = 0, \quad D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i \mid X_i] = 0. \quad (4)$$

Write $\ell_0(X) = \mathbb{E}[Y \mid X]$. Our partialling-out score uses ℓ , not the structural nuisance g :

$$\psi_i^{PLR}(\theta, \ell, m) = (D_i - m(X_i)) \{Y_i - \ell(X_i) - \theta(D_i - m(X_i))\}, \quad (5)$$

$$\hat{\theta} = \frac{\sum_i \hat{V}_i (Y_i - \hat{\ell}_i)}{\sum_i \hat{V}_i^2}, \quad \hat{V}_i = D_i - \hat{m}_i. \quad (6)$$

Neyman orthogonality follows because conditional treatment residuals have mean zero and the score's nuisance derivatives vanish at the truth. With heterogeneous effects, a residualized PLR coefficient is generally an overlap-weighted functional, not the unweighted ATE.

³We use the R calls `did::att_gt` and `aggte(type="simple")`.

⁴The latter uses `fixest::feols`.

⁵The R implementation uses `fixest::sunab` and `aggregate(..., "ATT")`. We use Sun–Abraham because the dCDH estimator could not be installed.

The interactive regression model defines $g_d(X) = \mathbb{E}[Y \mid D = d, X]$. Our ATE moment is

$$\begin{aligned} \psi_i^{IRM}(\theta, g, m) &= g_1(X_i) - g_0(X_i) + \frac{D_i(Y_i - g_1(X_i))}{m(X_i)} \\ &\quad - \frac{(1 - D_i)(Y_i - g_0(X_i))}{1 - m(X_i)} - \theta, \quad \mathbb{E}[\psi_i^{IRM}] = 0. \end{aligned} \quad (7)$$

For the empirical linear nuisance family, we use cross-validated Lasso for outcomes and cross-validated L2-penalized logistic regression for treatment probabilities.⁶ We fix the random-forest and gradient-boosting settings before estimation. We clip empirical propensity predictions at the saved threshold 0.01 and its upper complement. We average scores without normalized-IPW weights. We fit outcomes within treatment arms. Clipping is a numerical convention, not proof of overlap, and can affect the functional where support is weak.

A fixed partition $\{I_k\}_{k=1}^K$ supplies training complements I_k^c . For $i \in I_k$, we fit nuisances only on I_k^c and predict them on i . We use shuffled K-folds in simulations and common stratified folds across empirical PLR/IRM learner combinations. $K = 5$ with saved seed 20261007. We pool all out-of-fold scores for the coefficient, rather than averaging unrelated foldwise regression slopes. The in-sample orthogonal comparator deliberately predicts in-sample using the same orthogonal PLR algebra; it is not the nonorthogonal Lasso plug-in.

3.3 Inference, plug-in bias and held-out effects

For PLR define $J = \mathbb{E}[V^2]$ and influence $\phi_i = V_i(Y_i - \ell_i - \theta V_i)/J$. For simulation inference, we report

$$\widehat{SE}_{\text{sim}} = \frac{\text{sd}_{n-1}(\widehat{\phi}_i)}{\sqrt{n}}. \quad (8)$$

For empirical inference, we instead use the score second moment and Jacobian:⁷ $\widehat{SE}^2 = n^{-1} \overline{\psi_i^2} / \overline{\partial_\theta \psi_i}^2$. For IRM the Jacobian is minus one. We use analytic normal intervals for empirical inference and the specified normal cutoff for Monte Carlo coverage. For ordinary OLS, we use HC1 sandwich covariance, $(Z'Z)^{-1}(\sum_i Z_i Z_i' e_i^2)(Z'Z)^{-1}n/(n - \dim Z)$.

For the genuine Lasso plug-in, we solve

$$\min_{a, \theta, \gamma} \frac{1}{2n} \sum_i (Y_i - a - \theta D_i - X_i' \gamma)^2 + \lambda \|\gamma\|_1. \quad (9)$$

The intercept and treatment coefficient are unpenalized. We project Y and X off $[1, D]$, fit no-intercept Lasso on those residuals, then regress $Y - X'\widehat{\gamma}$ on intercept and treatment. Its coefficient condition uses D , rather than a residualized treatment against X ; nuisance shrinkage can therefore create first-order bias. The naive HC1 interval treats fitted nuisance coefficients as fixed. Penalized regressors in this plug-in use their projected native scale; sparse DML nuisances use training-fold standardization. This is not a pure cross-fitting-only comparison.

We assess latent-confounding sensitivity using the official DoubleML IRM definitions. With outcome residual variance σ^2 and squared Riesz norm ν^2 , the adversarial bias bound is

$$B = |\rho| \sqrt{\sigma^2 \nu^2 c f_y \frac{c f_d}{1 - c f_d}}, \quad \theta \in [\widehat{\theta} - B, \widehat{\theta} + B]. \quad (10)$$

Here $c f_y$ is the latent share of outcome residual variation; IRM $c f_d$ measures the relative gain in average conditional treatment precision, not a PLR treatment-residual R^2 . We set $\rho = 1$, both parameters to the Cartesian grid $\{.01, .03, .05, .10\}$, level95% and null zero. RV is the equal-strength

⁶The learners are `LassoCV` and `LogisticRegressionCV`; the latter retains its default L2 penalty.

⁷These are the analytic score-variance conventions used by DoubleML.

threshold bringing an effect bound to zero; RVa uses confidence bounds. Endpoint confidence limits are one-sided 95% and include estimated influence uncertainty for the bound.⁸ This model-based exercise is not proof of no confounding.

We train the forest on one random sample half and evaluate on 4958 held-out observations. We fit a local orthogonal residual-treatment moment with honest forest splitting.⁹ A separate training-half nuisance fit yields held-out AIPW scores Γ_i , equal to the IRM score before subtracting θ . We pre-specify for this diagnostic round four right-closed empirical income quantiles on the original held-out sample. Numeric cutoffs are computed empirically, not retrospectively pre-specified thresholds or forest-discovered groups. Income-quartile ATEs are means of Γ_i with sample-mean influence SEs. For the BLP, we regress Γ_i on an intercept and centered learned CATE, with HC1 conditional on training fits. Averaged individual forest interval endpoints are only descriptive, not group-mean confidence limits.

DiD inference is unit clustered for R TWFE/Sun–Abraham and analytic influence based for CS. For the county sensitivity calculation, we use the dynamic influence-function covariance and HonestDiD relative-magnitude bounds for the impact effect. We use the analytic influence-function covariance for this calculation.

4 Results

4.1 Two simulations and comparator consistency

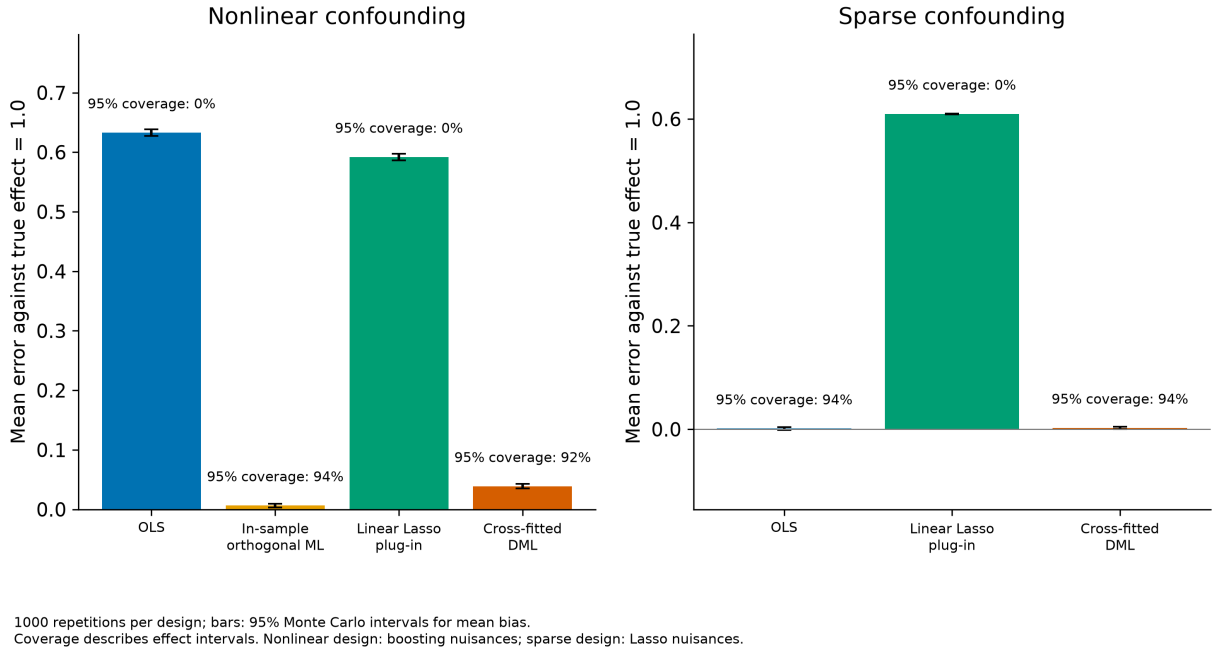


Figure 1: Biases across the original nonlinear and predeclared sparse designs. The original-design panel retains both the in-sample orthogonal estimator and the genuine linear Lasso plug-in. Error bars reflect Monte Carlo uncertainty.

In the original design, in-sample orthogonal bias is 0.007, versus 0.592 for the misspecified linear plug-in and 0.039 for cross-fitted DML. Its favorable historic naive performance is not removed. In the sparse design, plug-in bias is 0.610 and DML bias 0.003; DML RMSE is 0.040. Nonlinear DML coverage is 92 percent, while sparse-design DML coverage is 94 percent. These results illustrate the

⁸Definitions and analytic inference: <https://docs.doubleml.org/stable/guide/sensitivity.html>.

⁹The estimator is `CausalForestDML`.

Table 2: Estimator performance in both declared Monte Carlo designs

Design	Estimator	Bias	RMSE	MC SE	Coverage	Wilson 95%
Nonlinear	OLS	0.633	0.639	0.003	0.00	[0.0000, 0.0038]
Nonlinear	In-sample orthogonal ML	0.007	0.050	0.002	0.94	[0.9269, 0.9557]
Nonlinear	Linear Lasso plug-in	0.592	0.599	0.003	0.00	[0.0000, 0.0038]
Nonlinear	Cross-fitted DML	0.039	0.072	0.002	0.92	[0.9037, 0.9371]
Sparse	OLS	0.001	0.051	0.002	0.94	[0.9213, 0.9513]
Sparse	Linear Lasso plug-in	0.610	0.610	0.000	0.00	[0.0000, 0.0038]
Sparse	Cross-fitted DML	0.003	0.040	0.001	0.94	[0.9213, 0.9513]

Source: author’s calculations from the documented inputs. Coverage refers to nominal 95% treatment-effect intervals; MC SE is the standard error of mean bias across repetitions. Wilson intervals quantify uncertainty in estimated coverage, not treatment effects.

declared mechanisms rather than establishing an invariant ranking over learners and designs. The sparse design makes treatment strongly correlated with the active confounders. Residualizing Y and X off intercept and treatment leaves population active covariance 0.081633, below fixed native-scale Lasso penalty 0.132895. Shrinkage can omit that remaining confounding signal; unadjusted population omitted-confounder bias is 0.612245, near finite-sample plug-in bias 0.610. This is population intuition plus an executed finite-sample finding, not universal DML superiority; no penalty is changed.

4.2 Historical retirement-account replication

Table 3: Historical eligibility and net financial assets

Estimator	Nuisance	Estimate	SE	Lower 95%	Upper 95%
OLS	Linear	5896	1524	2909	8883
PLR	Lasso	6097	1464	3228	8966
IRM	Lasso	1279	4142	-6839	9398
PLR	RF	8833	1267	6350	11316
IRM	RF	8172	1213	5795	10549
PLR	GB	9196	1326	6597	11795
IRM	GB	7917	1156	5652	10183

Source: author’s calculations from the documented inputs. USD outcome units; PLR and IRM do not generally identify the same functional under heterogeneous effects. OLS uses HC1; PLR/IRM use score-based normal intervals.

The IRM random-forest estimate is \$8172, with interval [\$5795, \$10549]. This conditional interpretation assumes the measured covariates remove confounding and the relevant overlap conditions hold. Eligibility is not randomized, and the estimate is not an effect of actual participation or a prediction of a modern retirement-policy reform.

We report all learner/estimator combinations rather than selecting the largest coefficient or narrowest interval. PLR restricts the outcome relation and can weight heterogeneous effects differently from IRM. Numerical differences therefore reflect both nuisance fitting and the estimand, not merely estimator precision. Raw covariates are used even for the linear nuisance family; no more flexible polynomial replication has been silently substituted. RV is 8.411% and RVa 4.940% under the fixed confounding model. RF and GB refit estimates are unchanged; the Lasso estimate changes by \$0.000075423 because of numerical computation, disclosed in the exported comparison. These diagnostics preserve the target population.

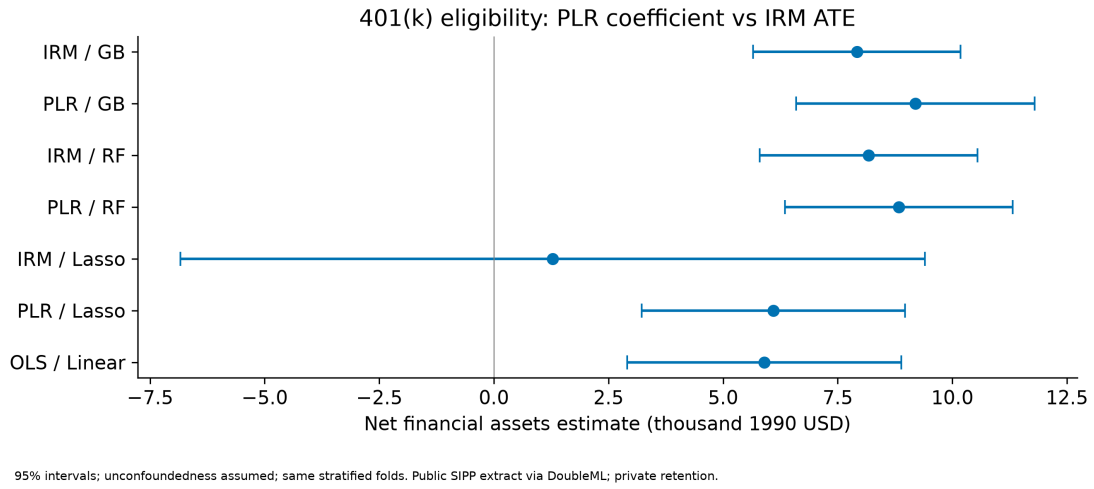


Figure 2: Stored eligibility estimates with score-based or HC1 intervals. Outcome is historical net financial assets; estimator family and learner are separate choices.

Table 4: Raw out-of-fold propensity clipping and arm-specific effective sample sizes

Learner	Arm	n	Clipped	Share	Removed	IPW ESS
Lasso	Control	6233	0	0.0000	0	4613.7
Lasso	Treatment	3682	0	0.0000	0	3023.3
RF	Control	6233	51	0.0082	0	5213.3
RF	Treatment	3682	3	0.0008	0	1130.5
GB	Control	6233	0	0.0000	0	5572.0
GB	Treatment	3682	0	0.0000	0	2422.9

Source: author calculations; thresholds 0.01 and 0.99. No observation trimming. Weights $1/m$ clipped for treatment, $1/(1-m \text{ clipped})$ controls. ESS: squared sum of weights divided by sum of squared weights. Raw-probability ESS also exported; no score normalization is implied.

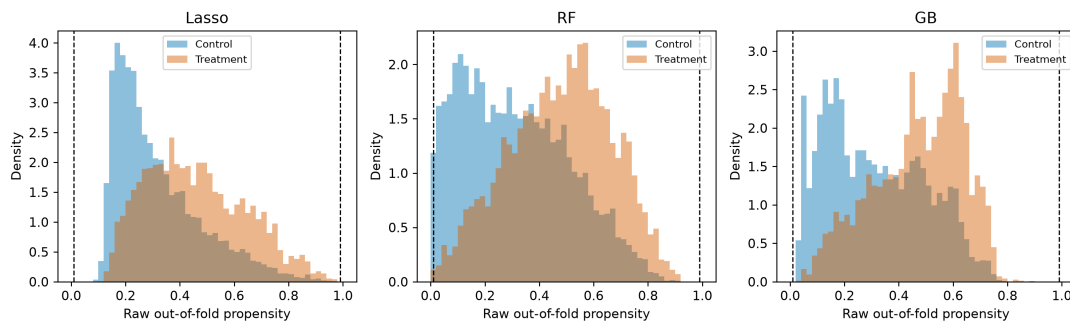


Figure 3: Treatment/control distributions of raw out-of-fold propensity predictions, before clipping, with both fixed thresholds marked. No observations are removed.

Table 5: RF-IRM sensitivity over the entire fixed confounding grid

cf y	cf d	Effect lower	Effect upper	Conf. lower	Conf. upper
0.01	0.01	7238	9107	5149	11153
0.01	0.03	6537	9808	4291	11984
0.01	0.05	6039	10306	3646	12614
0.01	0.10	5073	11272	2327	13909
0.03	0.01	6554	9791	4312	11963
0.03	0.03	5340	11005	2699	13543
0.03	0.05	4477	11867	1482	14744
0.03	0.10	2804	13541	-976	17185
0.05	0.01	6083	10262	3703	12557
0.05	0.03	4516	11829	1537	14690
0.05	0.05	3402	12943	-86	16300
0.05	0.10	1241	15103	-3338	19538
0.10	0.01	5217	11128	2528	13711
0.10	0.03	3001	13344	-682	16892
0.10	0.05	1426	14919	-3057	19257
0.10	0.10	-1630	17974	-7760	23952

Source: author calculations. $\rho = 1$, 95% one-sided confidence endpoints for effect bounds, null = 0, historical USD. $RV = 0.084107$, $RVa = 0.049404$. Parameters refer to latent outcome variation and the relative gain in average conditional treatment precision; they are not proof of absence of confounding.

Table 6: Staggered-adoption simulations with the same treated-cell ATT target

Estimator	Repetitions	Bias	RMSE	MC SE	Coverage
CS	1000	0.001	0.057	0.002	0.96
SunAbraham	1000	0.001	0.057	0.002	0.94
TWFE	1000	-1.056	1.060	0.003	0.00

Source: results/did_monte_carlo.csv; R simulation. Unit-cluster inference for TWFE/Sun-Abraham; analytic influence inference for CS. The Python Monte Carlo experiment is reported separately and is not pooled here.

Table 7: County teen-employment application

Estimator	Estimate	SE	Lower 95%	Upper 95%
TWFE	-0.0365	0.0133	-0.0625	-0.0105
CS_simple	-0.0418	0.0115	-0.0643	-0.0192

Source: author's calculations from the documented inputs. Log-employment units. CS conditions on log population; TWFE is unadjusted, so comparison also changes adjustment and aggregation.

4.3 Staggered adoption: simulated and empirical comparisons

The synthetic comparison aligns the treated-cell ATT target across methods. CS and Sun-Abraham handle cohort/event heterogeneity under the declared assumptions; conventional TWFE has different implicit comparisons. The county application additionally changes covariate adjustment between TWFE and CS. Differences are therefore not evidence that one coefficient is a clean causal benchmark for the other. Group-time cells and standard errors remain available in the aggregate results for a reader who wishes to inspect that aggregation. We aggregate retained group-time effects with constant cohort-size weights $w_g = n_g / (n_{2006} + n_{2007})$. Let $\phi_{i,g,e}$ be the retained influence function. Our reference-normalized estimate and influence are

$$\hat{\tau}_e = \sum_g w_g (\widehat{ATT}_{g,g+e} - \widehat{ATT}_{g,g-1}), \quad \phi_{i,e} = \sum_g w_g (\phi_{i,g,e} - \phi_{i,g,-1}), \quad \hat{V} = N^{-2} \sum_i \phi_i \phi_i'. \quad (11)$$

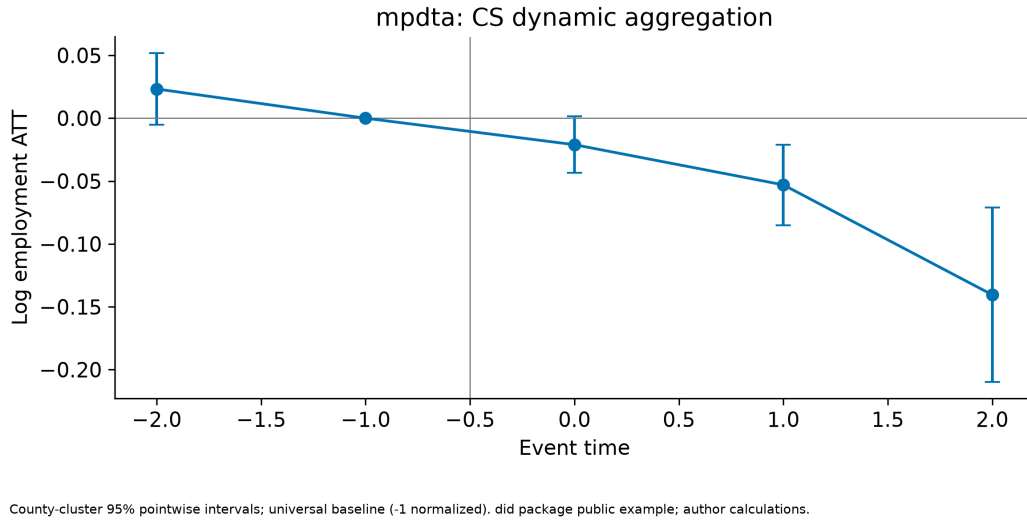


Figure 4: Saved county dynamic group-time effects, with the universal reference period. The limited pre-period information is retained rather than enlarged through a new event-window fit.

Table 8: Late-cohort, balanced-window comparison with reference event minus one

Event	Estimate	SE	Lower 95%	Upper 95%
-2	0.0232	0.0145	-0.0051	0.0516
-1	0.0000	0.0000	0.0000	0.0000
0	-0.0218	0.0125	-0.0464	0.0027

Source: author calculations. Cohorts 2006 and 2007, event times -2, -1 and 0; never-treated controls, unchanged population-adjusted $ATT(g, t)$, constant cohort-size weights. Different target from original dynamic analysis; its Hon-estDiD bounds are not transferred.

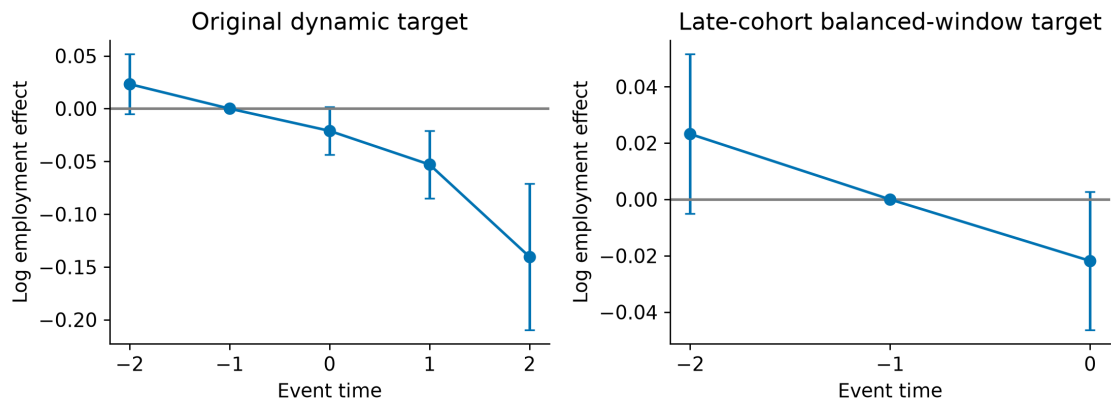


Figure 5: Original dynamic event study and late-cohort balanced-window comparison. Fixed cohorts 2006 and 2007 contribute at event times -2, -1 and 0, with -1 the reference. These panels have different estimands and cohort composition.

Never-treated controls and existing population-adjusted ATT specifications remain unchanged. The limited window changes the target; no balanced $[-2, +2]$ result is claimed. Existing HonestDiD bounds belong to the original dynamic target and are not transported to this comparison.¹⁰

4.4 Heterogeneity and sensitivity

Table 9: Held-out orthogonal income-quartile-group effects

Income quartile	Holdout n	ATE	SE	Lower 95%	Upper 95%
1	1241	5601	1532	2598	8604
2	1238	2778	1171	483	5072
3	1240	8469	2681	3214	13724
4	1239	19639	6433	7030	32248

Source: author’s calculations from the documented inputs. Groups are income quartiles of the held-out sample; they are not forest-ranked GATES. Influence intervals condition on training-half nuisance fits.

The held-out calibration slope is 1.077, with interval $[0.308, 1.847]$. Calibration uses the same held-out AIPW outcome as the group analysis and conditions on the training-half forest/nuisance fits. It checks association between learned heterogeneity and orthogonal effects; it does not prove unconfoundedness or transportability. Income groups are not ranks of the estimated forest score and are not labelled forest-ranked GATES. The sensitivity bounds already include zero at the

Table 10: Sensitivity of the impact-event ATT to relative pretrend violations

M-bar	Lower	Upper	Method	Restriction
0.0	-0.0430	0.0012	C-LF	DeltaRM
0.5	-0.0545	0.0163	C-LF	DeltaRM
1.0	-0.0720	0.0352	C-LF	DeltaRM
2.0	-0.1157	0.0803	C-LF	DeltaRM

Source: results/honestdid.csv; HonestDiD relative-magnitude restriction, impact contrast, nominal 95% bounds. Sparse usable leads limit informativeness.

strictest saved relative-magnitude setting. Sparse non-reference leads limit the empirical content of the sensitivity exercise. A conventional post-treatment confidence interval and a sensitivity set allowing violations of parallel trends answer different questions. We report the entire predeclared grid, including values that weaken the finding; no threshold was selected to maintain significance.

We compute the underlying covariance as the saved dynamic influence-function cross-product divided by the square of the unit count. The event immediately before treatment is omitted as the normalized reference, and the contrast selects the impact effect. Retained estimator objects remain private; the paper provides aggregate bounds only.

5 Robustness

The two declared designs intentionally change dimension, nuisance family and confounding shape. Their contrast illustrates a theoretical vulnerability of regularized plug-in estimation and cannot isolate the effect of sample splitting alone. The original in-sample orthogonal comparator remains a distinct row in the table, with identical labelling in the figure and interpretation.

¹⁰The documented `balance_e` criterion concerns post-treatment exposure: <https://bcallaway11.github.io/did/reference/aggte.html>.

Table 11: Alternative specifications and inference

Declared alternative	Specification	Interpretation
DML comparators	OLS; in-sample orthogonal ML; genuine Lasso plug-in; cross-fitted DML	Both original and sparse designs retained
Nuisance families	Linear/Lasso, random forest, gradient boosting	PLR and IRM side-by-side; common folds
DiD estimator	TWFE; CS; Sun–Abraham	Sun–Abraham; dCDH could not be installed
DiD adjustment	Unadjusted TWFE; population-adjusted CS	Not a like-for-like estimator-only contrast
Sensitivity	HonestDiD relative-magnitude grid	Impact contrast and saved covariance
Heterogeneity	Forest averages; held-out AIPW groups; BLP	Pointwise forest bounds are not group intervals
Dose estimand	ATT(d) versus ACRT(d)	Known synthetic illustration only
Data preprocessing	Raw covariates; native dtype restoration	No winsorization, row deletion or polynomial expansion
Overlap / inference	Fixed propensity clipping; analytic score SEs	No clipping sensitivity or repeated-split run performed

Source: comparative estimates under the stated nuisance and inference specifications.

The fixed overlap/sensitivity grid is reported, but no clipping-grid, repeated-split or outcome-transformation sensitivity is added. We use Sun–Abraham because the dCDH estimator could not be installed. Inference choices, data restrictions and design additions are declared rather than retrofitted to the headline. The complete tables include methods that outperform DML on one metric and alternatives that fail to cover the known effect.

We compare estimators on the same 1000 completed seeded repetitions within each experiment. Seed sequences and draw specifications are held fixed, and incomplete or failed repetitions are excluded consistently across estimators. Coverage uncertainty remains finite. We use Wilson intervals even at zero successes, avoiding the misleading certainty suggested by a zero plug-in binomial SE. The appendix records uncertainty and the exact DGP functions.

6 Limitations and future work

IPW ESS is a weight-concentration diagnostic; it does not include propensity-estimation uncertainty and is not a measure of causal validity. Orthogonality reduces sensitivity to nuisance estimation under rate, regularity and identification conditions; it does not repair omitted confounders, reverse causality, interference or lack of common support. In the historical application, some covariates are contemporaneous, so a claim that all controls precede eligibility is unwarranted. The observational assumptions remain the principal limit on causal interpretation.

Both synthetic designs are modest, finite simulations. Their learner families, dimensions and nuisance scaling differ. The genuine Lasso plug-in is deliberately evaluated with naive nuisance-fixed inference, while the historic in-sample orthogonal comparator uses a different score. Neither is a universal stand-in for all naive machine learning. The sparse design was added to illustrate theory before its draws, not to tune results; the retained nonlinear panel visibly qualifies the broader message.

The county illustration has sparse leads, limited calendar support and distinct adjustment across estimators. HonestDiD bounds should not be interpreted as verification of parallel trends. Sun–Abraham addresses heterogeneous event-time effects; it does not implement the dCDH estimator. Forest calibration and group inference are conditional on the training-half fits; one sample split supplies limited evidence on split sensitivity.

Package-level convergence details and learner warnings were not exhaustively exported into the saved aggregate results. Their absence is marked as unreported rather than asserted successful. Source data licence scope permits the analysis described here with private retention and does not authorize redistribution. The county illustration does not establish effects for a modern German minimum-wage reform.

7 Conclusion

The laboratory demonstrates why a treatment-effect procedure must be described by its estimand and moment, not simply by the phrase “machine learning.” Sparse regularization creates severe plug-in bias in the declared design, while an orthogonal cross-fitted estimator performs well. The original nonlinear design preserves a favorable in-sample orthogonal result, preventing an overgeneralized cross-fitting narrative.

The empirical and panel examples extend the same discipline: state unconfoundedness or parallel trends, preserve all declared comparators, report inference appropriate to the score, and distinguish descriptive heterogeneity from formal group effects. We report the estimates together with the identifying assumptions and limits of each application. Future work includes matched-learner/scaling/oracle factorials, correlated and denser or $p \geq n$ designs, fuller RMSE/mean-SE Monte Carlo uncertainty, score concentration and observed-confounding benchmarks, repeated splits and DAG-based control sensitivity, wealth-outcome/transportability checks, income-group contrasts and forest-ranked groups, matched-covariate DiD comparisons, balanced-target sensitivity and pretrend-power studies, and observational continuous-dose designs. These extensions are outside the scope of the present study.

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A Implementation and reproducibility

We implement PLR moments and HC1 inference in `core.py`; the partially penalized Lasso and sparse design in `sparse.py`; DoubleML and held-out forest/AIPW estimators in `pipeline.py`; and staggered-adoption, sensitivity and decomposition estimators in `scripts/did.R`. Environment locks record package versions and default settings.

B Monte Carlo design and estimator definitions

We use the following nonlinear DGP:

$$X_i \sim N(0, I), \quad U_i, V_i \sim N(0, 1), \quad (12)$$

$$D_i = \frac{1}{2}X_{i1}^2 + \sin(X_{i2}) + V_i, \quad (13)$$

$$Y_i = D_i + X_{i1}^2 + \cos(X_{i2}) + \frac{1}{2}X_{i4} + U_i. \quad (14)$$

The original design uses $n = 500$, $p = 20$, with 1000 repetitions; gradient boosting uses 60 trees of maximum depth 2. A linear nuisance is misspecified. The sparse DGP uses $n = 600$, $p = 200$, 5 active independent Gaussian coordinates:

$$D_i = 1.5 \sum_{j=1}^5 X_{ij} + V_i, \quad Y_i = D_i + \sum_{j=1}^5 X_{ij} + U_i. \quad (15)$$

The true effect is 1, with 1000 repetitions and seed 20261008 plus the repetition index. The fixed sparse Lasso penalty is $\sqrt{2 \log p/n} = 0.132895$. Sparse DML standardizes within each training fold; OLS controls all candidate coordinates; the joint partially penalized plug-in uses its native projected scale. The sparse solver cap is 10000 iterations with tolerance 1e-07; the saved aggregates do not expose per-fit convergence paths.

The staggered synthetic panel has $G_i \in \{0, 3, 5, 7\}$, with treatment absorbing at adoption. Its known effect is

$$\tau_{it} = D_{it}(t - G_i + 1)\{1 + 0.7 \mathbf{1}(G_i = 5) + 1.5 \mathbf{1}(G_i = 7)\}, \quad Y_{it} = \alpha_i + 0.15t + \tau_{it} + e_{it}. \quad (16)$$

The Python and R engines use sample sizes 400 and 300, respectively, with different random generators; their draws are not pooled. Within each R draw, all estimators use the same treated-cell truth. Bias is the mean estimation error; RMSE is its quadratic mean; bias MC SE is the error sample SD divided by the square root of repetitions. Coverage tests the known truth against each reported normal interval. With R completed repetitions, coverage MCSE is $\sqrt{\hat{c}(1 - \hat{c})/R}$; Wilson intervals accompany it. Zero plug-in MCSE at coverage zero or one is not certainty.

Table 12: Monte Carlo uncertainty and comparator labels

Design	Estimator	Coverage MC SE	Bias MC SE
Nonlinear	Cross-fitted DML	0.0085	0.0019
Nonlinear	In-sample orthogonal ML	0.0073	0.0016
Nonlinear	Linear Lasso plug-in	0.0000	0.0029
Nonlinear	OLS	0.0000	0.0028
Sparse	Cross-fitted DML	0.0076	0.0013
Sparse	Linear Lasso plug-in	0.0000	0.0004
Sparse	OLS	0.0076	0.0016

Source: author’s calculations from the documented inputs. A zero binomial plug-in SE at zero coverage is not certainty: use the nonzero Wilson upper endpoint in the main table.

C Event diagnostics, decomposition and dose estimands

The Goodman–Bacon decomposition checks that weighted components reconstruct the unadjusted TWFE coefficient. Positive comparison weights do not eliminate problematic already-treated comparisons or guarantee positive implicit weights on heterogeneous treatment effects. Unit-root tests, VAR lag diagnostics and

Table 13: DiD Monte Carlo coverage uncertainty at actual completed counts

Engine	Estimator	R	Coverage	MCSE	Wilson 95%
Python	CS_unconditional	1000	–	–	[–,–]
Python	TWFE	1000	0.0000	0.0000	[0.0000,0.0038]
R	CS	1000	0.0064	0.0064	[0.9426,0.9679]
R	SunAbraham	1000	0.0072	0.0072	[0.9291,0.9575]
R	TWFE	1000	0.0000	0.0000	[0.0000,0.0038]

Source: author simulations. Python CS has no interval in its original implementation, so coverage is unreported. Zero endpoint MCSE is not certainty; engines use separate seeded draws.

Table 14: Included cohorts, fixed weights and event support

cohort	n	weight
2006.0	40.0	0.233918128654971
2007.0	131.0	0.766081871345029

Source: author calculations; support matrix is exported before aggregation. Full row-level influence functions remain private.

Table 15: Cohort-by-event-time support before balanced aggregation

cohort	event	time	observed_calendar	group_time_available	reference	balanced_included
2004	-2	2002	False	False	False	False
2006	-2	2004	True	True	False	True
2007	-2	2005	True	True	False	True
2004	-1	2003	True	True	True	False
2006	-1	2005	True	True	True	True
2007	-1	2006	True	True	True	True
2004	0	2004	True	True	False	False
2006	0	2006	True	True	False	True
2007	0	2007	True	True	False	True
2004	1	2005	True	True	False	False
2006	1	2007	True	True	False	False
2007	1	2008	False	False	False	False
2004	2	2006	True	True	False	False
2006	2	2008	False	False	False	False
2007	2	2009	False	False	False	False

Source: author calculations; non-existing event cells are unsupported, not fabricated. The balanced comparison uses only cohorts 2006 and 2007 at -2, -1 and 0. balance_e concerns post-treatment exposure, not balanced pre-treatment support.

Table 16: Saved dynamic DiD estimates and sparse pre-period evidence

Event time	Estimate	SE	Lower 95%	Upper 95%
-2	0.0232	0.0145	-0.0051	0.0516
-1	0.0000	–	–	–
0	-0.0211	0.0115	-0.0436	0.0015
1	-0.0530	0.0163	-0.0850	-0.0210
2	-0.1404	0.0354	-0.2098	-0.0711

Source: author's calculations from the documented inputs. The normalized reference period has no estimated interval. All saved leads/lags are shown; no extra event window was fitted.

Table 17: Goodman–Bacon accounting by comparison class

Comparison class	Weight sum	Contribution
Earlier vs Later Treated	0.0833	-0.0016
Later vs Earlier Treated	0.0539	0.0002
Treated vs Untreated	0.8628	-0.0351

Source: results/bacon.csv; contributions are weighted component estimates, not new regressions. Already-treated comparisons remain visible. Positive component weights do not rule out negative implicit treatment-effect weights.

IRF bootstraps are not applicable to these cross-sectional/panel examples. Learner convergence histories are unreported in the saved results.

For the independent dose illustration, $X \sim U[0, 1]$, $D = 0.2 + 0.6X + U[0, 0.2]$, and $a = 1 + 2X$. Known outcomes obey

$$Y(d) - Y(0) = ad + \frac{1}{2}d^2, \quad \partial_d Y(d) = a + d. \quad (17)$$

The reported dose-group ATT averages the known effect at realized dose, while ACRT averages its known derivative. Selection on gains makes them different. These descriptive synthetic quantities are not a continuous-dose DiD estimate from observational data.

Table 18: Known effects and derivatives in the dose-selection illustration

Dose quartile	n	Mean dose	ATT	SE(ATT)	ACRT	SE(ACRT)
1	1250	0.367	0.553	0.004	1.665	0.007
2	1250	0.531	1.090	0.006	2.305	0.008
3	1250	0.683	1.804	0.007	2.971	0.008
4	1250	0.840	2.643	0.009	3.556	0.006

Source: author’s calculations from the documented inputs. Descriptive means of known synthetic potential outcomes, with sample-mean SEs; no causal effect has been estimated from observational dose variation.